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Analysis of laser welding on dissimilar materials with the influence of adding interlayer



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ABSTRACT

The laser welding of dissimilar materials poses several challenges in practical applications, such as molten pool instability, the formation of intermetallic compounds (IMCs), and cracking caused by differential thermal stresses. In this study, we investigated the impact of incorporating a nickel (Ni) foil as an intermediate layer on lap joints of dissimilar materials (SS304 and Al6061) welded using the continuous laser welding (CLW) technique. The thermomechanical simulation accurately predicted the width and depth of SS304–Ni–Al6061 joints, with deviations of merely 3.92% and 19.04% from experimental results, respectively. Residual stress analysis revealed that the addition of a nickel foil reduced residual stress in the heat-affected zone (HAZ) and improved joint fatigue life. Furthermore, the incorporation of 0.1 and 0.2 mm Ni foils enhanced joint strength and effectively suppressed the formation of brittle intermetallic compounds, resulting in a remarkable maximum lap shear force of up to 1026 N. The simulation results for the shear load-displacement curve displayed excellent agreement with experimental data, with a deviation of only 6%. Notably, joints with the Ni foil demonstrated a greater ability to prevent crack formation caused by thermal expansion mismatch compared to joints without the Ni foil. The Ni interlayer served as a diffusion barrier and stress buffer, promoting a more uniform microstructure and reducing the likelihood of crack formation in bonded materials. Overall, this study comprehensively investigates the impact of incorporating a nickel interlayer on laser-welded lap joints of dissimilar materials. The findings offer valuable insights for practical applications, enabling improved joint strength, reliability, and crack resistance in laser welding dissimilar materials.

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1. Introduction

Laser beam welding is widely regarded as a highly preferred method for the fusion joining of dissimilar metals such as stainless steel and aluminum. This technique offers

remarkable mechanical properties, including exceptional specific strength, excellent formability, superior corrosion resistance, and lightweight characteristics. As a result, it finds extensive applications in various industries, including automotive, aerospace, and numerous other sectors [1]. Continuous Laser Welding (CLW) is favored due to its ability to weld

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thicker materials, minimize heat-affected zones, and reduce residual stress post-welding. This technique is widely preferred in various industries owing to its advantages in terms of limiting the heat impact on the surrounding area and ensuring minimal residual stress upon completion of the welding process [2]. The welding of stainless steel and aluminum poses significant challenges due to their dissimilar chemical, physical, and thermodynamic properties. The inherent difficulties in achieving a sound weld between these metals arise from their poor weldability, forming brittle intermetallic compounds (IMCs) near the interface. These IMCs can compromise the mechanical properties of the welded joint and increase the likelihood of cracking [3]. The metallurgical issue of spatter and its impact can be enhanced by employing high welding speeds. Additionally, laser welding, with its stable laser penetration (keyhole) during the welding process and excellent controllability, offers a means to effectively suppress the formation of intermetallic compounds (IMCs) [4].

Chen et al. [3] conducted a study utilizing laser penetration overlap welding to compare the characteristics of joints with and without the inclusion of a nickel foil interlayer. Their findings demonstrated that the joint incorporating the nickel foil interlayer exhibited notably superior mechanical properties and toughness compared to the joint without it. However, the study primarily concentrated on microstructural analysis, the formation of intermetallic compounds (IMCs), and the assessment of mechanical properties. In Refs. [5,6], researchers have investigated the highly resistant ability of $\text{Yb}_4\text{Hf}_3\text{O}_{12}$ ceramic against the infiltration of molten CFAS at 1400 °C for varying periods, analyzing the formation of new phases and the barrier effect of the apatite reaction layer during the thermal corrosion process. Xu et al. [7] investigated the influence of different thicknesses of nickel foil on the performance of joints during the welding of steel and aluminum. Their study revealed that the thickness of the nickel foil exerts a substantial influence on the performance of the joints, especially their mechanical properties. In a separate study, Zhou et al. [8] aimed to join dual-phase stainless steel and 6016 aluminum alloy by incorporating different manganese (Mn) or silicon (Si) powders to enhance penetration depth. However, the addition of powders can present challenges in terms of control, and the agglomeration of powders can adversely affect the performance of the compounds, thereby introducing experimental errors. Yang et al. [9] also investigated the performance of weld joints using a nickel (Ni) coating. However, their study only demonstrated a modest 25% improvement compared to joints without a Ni coating. One limitation of their study was the absence of numerical simulations to facilitate convenient and efficient predictions, thereby reducing the need for extensive trial-and-error experimentation.

Having a thorough understanding of the thermal phenomena that occur during welding is crucial for accurately setting technical parameters, ensuring process stability, and achieving the desired joint quality [10–12]. As a result, researchers are continually searching for numerical models that accurately depict the distribution of energy fluxes, while considering real-world conditions [13,14]. These models play a pivotal role in designing and optimizing welded structures

and process parameters, ultimately saving time and costs and generating further interest in the field of welding fabrication processes [15,16]. However, the utilization of an interlayer for joining two dissimilar materials as a three-material model employing the finite element (FE) approach remains uncommon and limited.

The present research aims to investigate the effectiveness of a Ni foil interlayer with varying thicknesses (0.1 or 0.2 mm) in mitigating the formation of brittle intermetallic compounds (IMCs) at the interface of dissimilar materials, specifically SS304 and Al6061. To achieve this objective, a combination of simulations and experiments is employed to evaluate the interlayer's impact on various parameters such as melt pool interface width, melt pool penetration depth, thermal distribution of diffusion zones, and laser joining strength. The researchers implemented a fully coupled thermo-mechanical model using finite element (FE) analysis to simulate the behavior of the interlayer. Additionally, electron dispersive spectroscopy (EDS) was utilized to analyze the microstructures at the IMCs. Finally, the simulation results were validated by comparing them with experimental results obtained from lap laser-welded joints.

The novelty of this study lies in its focus on utilizing a Ni foil interlayer to address the challenges associated with dissimilar material welding, particularly in suppressing the formation of brittle IMCs. Moreover, the employment of a fully coupled thermo-mechanical model and EDS analysis provides a comprehensive understanding of the interlayer's impact on laser joining strength and the formation of microstructures. In summary, the findings of this study hold significant potential for the advancement of dissimilar material joining techniques.

2. Materials and methods in the experiment

2.1. Experimental material

The experimental materials employed in this study consisted of SS304 plates with dimensions of $100 \times 10 \times 1$ mm and Al6061 plates with dimensions of $100 \times 10 \times 1$ mm, which were prepared for laser beam welding. As an interlayer material, a Ni foil composed of Ni-99.9% alloy was chosen. The Ni foil had dimensions of $20 \times 10 \times 0.1$ mm or $20 \times 10 \times 0.2$ mm, depending on the experiment, and was positioned between the stainless steel and aluminum plates, as depicted in Fig. 1. Before welding, both SS304 and Al6061 materials were thoroughly cleaned with acetone to eliminate any oxide film on their surfaces. The Ni foil was stored in a warm and dry environment to minimize moisture absorption. It was then firmly clamped in place using a welding clamp to minimize the gap between the two welding materials, as shown in Fig. 2, while the flow chart of the specific experimental steps can be found in Fig. 3.

The lap joints were produced using a single continuous laser welding (CLW) process with a high-power YAG laser (Model number: JBL-LWF series). The laser possessed a maximum power of 3000 W and a maximum scanning speed of 5000 mm/s. During the welding process, the laser head focal distance was set at 25 mm above the surface of the plate, as

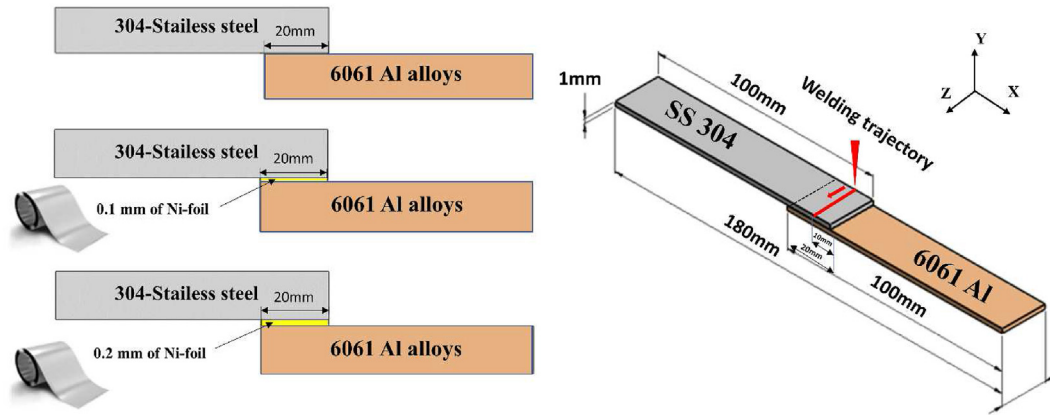


Fig. 1 – Schematic illustration of lap joint welding dimension.

illustrated in Fig. 2, and the laser beam diameter was 50 μm . To ensure data accuracy, each parameter set of laser power and scanning speed was repeated three times in separate experiments. Notably, the welding trials were conducted without the use of shielding gas in this particular study. Table 1 provides an overview of the experimental parameter sets used for laser welding and the lap joint samples. These parameter sets were optimized to achieve high-quality laser welding and lap joint samples, with a specific emphasis on producing robust and reliable joints.

2.2. Microstructural analysis and mechanical testing

For microstructural analysis, the specimens were sectioned into small metallographic samples perpendicular to the construction direction (XZ plane) using a water-cooled cut-off machine. These samples were then embedded in epoxy resin, followed by grinding with silicon carbide sandpaper up to a grit size of P240–P3000. Subsequently, diamond suspension was employed for polishing the samples, achieving a sizing of 0.3 μm . To ensure the elimination of surface debris and contaminants, the polished samples underwent a cleaning process with ethanol. Phase analysis was performed using scanning electron microscopy (SEM), optical microscopy (OM),

and energy-dispersive spectroscopy (EDS) utilizing a ZEISS Supra 55 instrument.

To measure the tensile shear load of various lap joint configurations, tensile shear tests were conducted using a Shimadzu AG-I testing machine equipped with a 2 kN load cell. The tests were conducted at room temperature with a testing speed of 1.5 mm/min. For each experimental parameter, five specimens of each lap joint configuration were tested to determine the tensile shear force and observe the fractured surfaces.

3. Numerical simulation

For numerical simulations, specific material data for each solver are required. The materials utilized in this study, namely SS304, Al6061, and nickel alloy, possess temperature-dependent properties. The thermal-physical properties of these materials are provided in Tables 2–5 [17–19]. Additionally, latent heat associated with the solid-liquid phase transition is considered and presented in Table 5 [18,19].

The Abaqus software was employed for investigating and simulating the laser welding process involving three different materials: stainless steel, aluminum alloy, and nickel alloy.

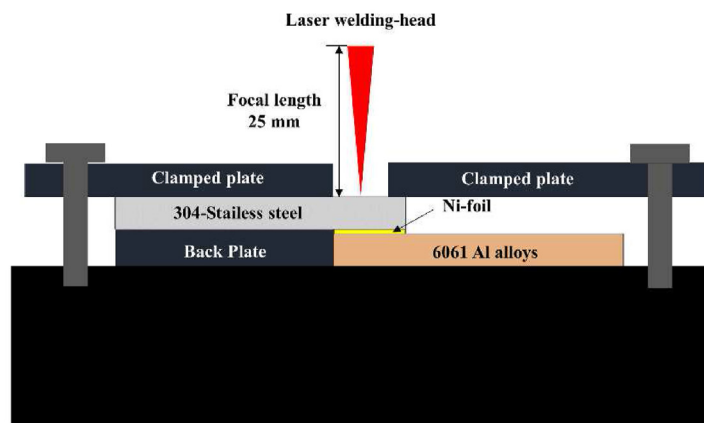
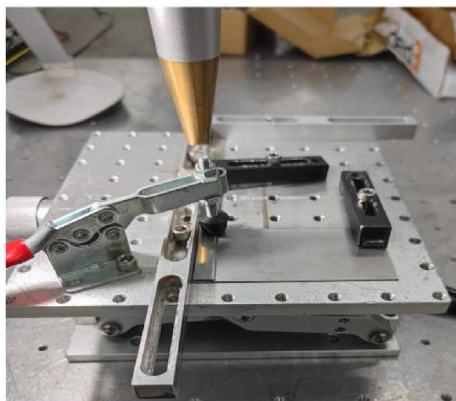


Fig. 2 – Schematic illustration of clamping setup for weld samples.

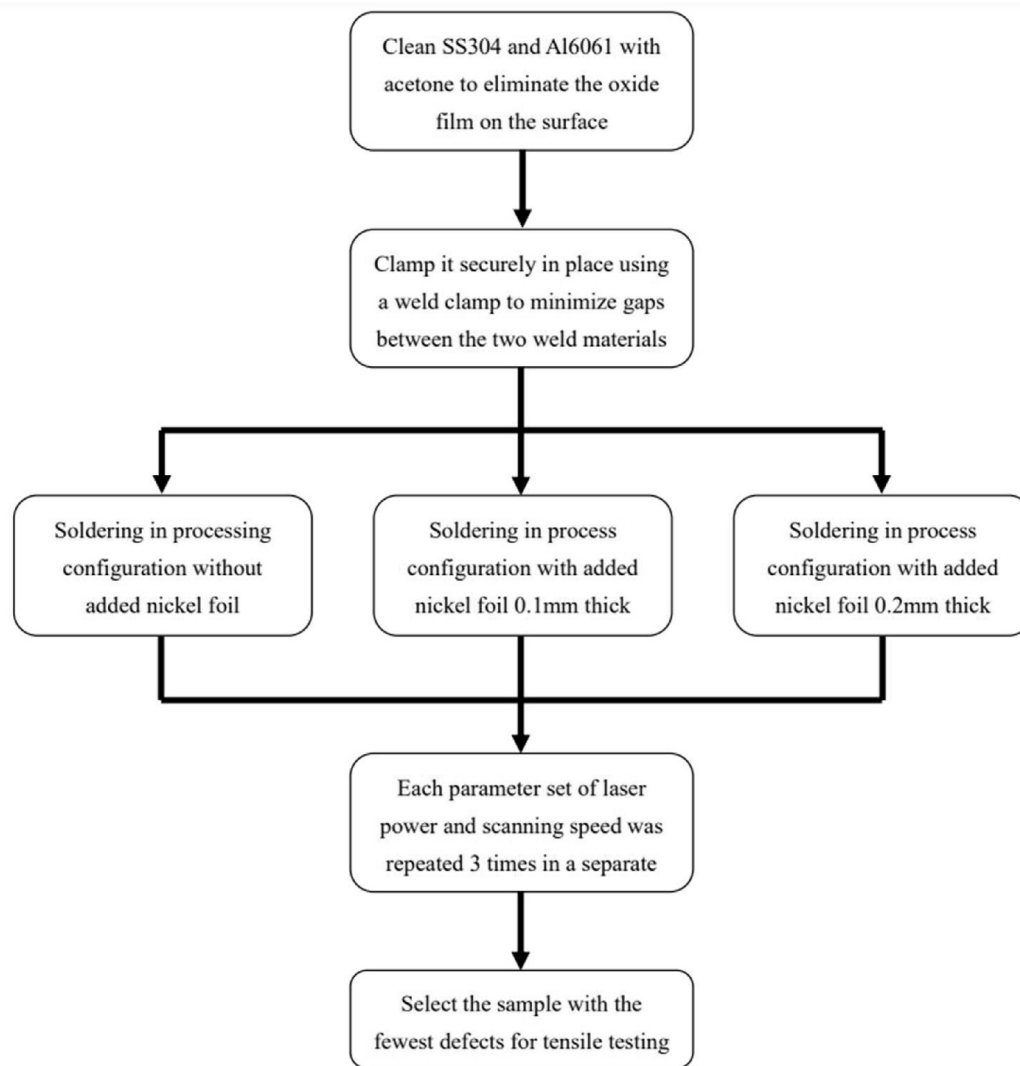


Fig. 3 – Flow chart of the experimental procedure.

Abaqus is a versatile software capable of handling various heat transfer problems, including uncoupled heat transfer, sequentially coupled thermal stress analysis, and fully coupled thermal stress analysis. With its comprehensive features, Abaqus can effectively model and analyze the intricate heat transfer and stress interactions during the laser

welding process for the aforementioned materials [20]. In sequential coupled thermal-stress analysis, a pure heat transfer problem is computed first, and the resulting data is then utilized as input for a subsequent stress and deformation analysis [20]. The fully coupled thermal-stress analysis is used when the thermal and mechanical properties greatly influence each other. The method computes the temperature and deformation field in the same analysis, thus yielding both thermal and mechanical results at the same time and eliminating the need for two separate analyses.

To simulate a moving heat source, the Abaqus subroutine Dflux is executed. This subroutine allows users to customize the functionality of Abaqus and employ more sophisticated analysis techniques. The Dflux subroutine is implemented using Fortran, which is the standard programming language utilized in Abaqus and well-suited for numerical computations of heat flux concerning both position and time [20].

As depicted in Fig. 4, the finite element (FE) analysis procedure for the coupled thermomechanical simulation consisted of three main steps: analysis of the thermal welding

Table 1 – Welding parameter set.

Iterations No.	Power (W)	V (m/min)
1	1800	1.5
2	1950	1.5
3	2100	1.5
4	2250	1.5
5	2400	1.5
6	2000	1.75
7	2100	2
8	2200	2.25
9	2300	2.5
10	2400	3

Table 2 – Properties of SS 304 alloy [17,18].

T (°C)	E (GPa)	N (No unit)	σ_y (MPa)	α^T (1/°C)	C (J/kg°C)	λ (W/m°C)	ρ (kg/m ³)
0	198	0.294	265	1.7 e–5	462	14.6	7900
100	193	0.295	218	1.74 e–5	496	15.1	7850
200	185	0.301	186	1.8 e–5	512	16.1	7800
400	167	0.318	170	1.86 e–5	525	18	7790
600	159	0.326	155	1.91 e–5	540	20.8	7780
800	151	0.333	149	1.96 e–5	577	23.9	7750
1000	100	0.336	91	2.02 e–5	604	28	7700
1200	60	0.339	25	2.07 e–5	676	33.7	7680
1400	15	0.36	21	2.11 e–5	692	39	7670
1500	10	0.388	10	2.16 e–5	700	70	7650

T - temperature; E – Young modulus; ν - Poisson ratio; σ_y - yield stress.

α^T - thermal expansion coefficient c - specific heat; λ -conductivity; ρ -density.

procedures, mechanical analysis, and cooling. Subsequently, a new mechanical analysis was initiated to estimate the tensile shear force.

3.1. Governing equation and boundary condition

During a thermomechanical process, it is necessary to elucidate the issues related to heat transfer through the conduction mode. The simulations are conducted using Lagrange coordinates, wherein the center coordinate of the heat source is defined as $x = v \cdot t$ for each time increment t [s], considering the welding speed v [m/s]. The heat input can be represented as the heat flux exerted on the material [17], which can be constructed using Eq. (1). The differential governing equation for the temperature field in the Abaqus software is based on the energy conservation equation incorporating Fourier's law [19]:

$$\int_V \rho \frac{\partial U}{\partial t} dV + \int_V \frac{\partial \delta T}{\partial z_\alpha} \left[\lambda \frac{\partial T}{\partial z_\alpha} \right] dV = \int_V \delta T q_v dV + \int_S \delta T q_s dS \quad (1)$$

where $\lambda = \lambda(T)$ [W/m°C] is thermal conductivity, $U=U(T)$ is internal energy [J/kg], q_v is laser beam heat source [W/m³], q_s is a boundary heat flux [W/m²], δT is a variational function, ρ is a density [Kg/m³], and $T = T(x_\omega, t)$ is temperature [°C].

Furthermore, heat losses from the workpiece to the surrounding ambient air are considered, taking into account convection, radiation, and evaporation [21]. The heat loss resulting from convection follows Newton's law of cooling, which can be constructed by Eq. (2) as

$$q_c = -h(T - T_0) \quad (2)$$

where q_c is the heat loss due to conduction and h is the heat transfer coefficient. T is the current temperature of the workpiece and T_0 is the ambient temperature. Heat loss due to radiation is given by Eq. (3) as

$$q_r = \epsilon \sigma (T^4 - T_0^4) \quad (3)$$

where ϵ is the emissivity factor for the material and σ is the Stefan-Boltzmann constant. The thermal strain of the material induced by heat can be computed by Eq. (4) as

$$\epsilon_{th} = \beta(T - T_0) \quad (4)$$

where β is the coefficient of thermal expansion. The mechanical analysis, conducted within the elastic-plastic range, relies on the classic equilibrium equations. These equations are supplemented by constitutive relationships and initial and boundary conditions, aiming to establish the external static determination of the relevant system [21]. The governing equations can be expressed as follows:

$$\nabla_\alpha \dot{\sigma}(x_\alpha, t) = 0, \dot{\sigma} = \dot{\sigma}^T \quad (5)$$

$$\dot{\sigma} = D_\alpha \dot{\epsilon}^e + \dot{D}_\alpha \epsilon^e \quad (6)$$

$$\sigma(x_\alpha, t_0) = \sigma(x_\alpha, T_s) = 0, \epsilon^e(x_\alpha, t_0) = \epsilon^e(x_\alpha, T_s) = 0 \quad (7)$$

$$\epsilon^T = \epsilon^e + \epsilon^p + \epsilon^{Th} \quad (8)$$

Table 3 – Properties of Al 6061 alloy.

T (°C)	E (GPa)	N (No unit)	σ_y (MPa)	α^T (1/°C)	C (J/kg °C)	λ (W/m°C)	ρ (kg/m ³)
0	70	0.331	276	2.18E-05	986	152	2781
100	66	0.332	260	2.38E-05	1094	167	2756
200	63	0.334	220	2.56E-05	1151	178	2695
300	58	0.340	90	2.72E-05	1194	183	2650
400	50	0.357	30	2.82E-05	1240	193	2610
500	36	0.393	10	2.96E-05	1293	210	2605

Table 4 – Properties of Ni alloy [19].

T (°C)	E (GPa)	N (No unit)	σ_y (MPa)	αT (1/°C)	C (J/kg °C)	λ (w/m°C)	ρ (kg/m ³)
0	206	0.305	92	1.23E-05	432	70	8900
100	200	0.307	84	1.31E-05	472	67	8870
200	195	0.308	78	1.36E-05	515	62	8830
400	183	0.310	65	1.44E-05	530	56	8780
600	170	0.312	46	1.59E-05	536	60	8750
700	162	0.313	36	1.63E-05	550	62	8700
800	154	0.314	27	1.67E-05	557	64	8660
1000	129	0.315	26	2.21E-05	584	69	8610

Table 5 – Latent heat is associated with the solid-liquid phase [15,18,19].

Nomenclature	SS304	Al6061	Ni
T_s (°C) -Solidus temperature	1400	581.85	1260
T_L (°C) -Liquidus temperature	1454	651.85	1336
H (J/kg)-Latent heat	260,000	290,000	292,000

where $\sigma = \sigma(\sigma_{ij})$ is stress tensor [N/m²], $\mathbf{x}_\alpha$ describes the position of the material particle point under consideration, (ϕ) is an inner exhaustive product, and $D = D(T)$ is a tensor of the temperature-dependent properties of the material. The elastic deformation is modeled using an isotropic law of Hooke while the plastic deformation is calculated using a plastic flow model complying with the Huber-Misses plasticity condition [22], where ϵ^T is the total strain is defined as a sum of elastic ϵ^e , plastic ϵ^p , and thermal ϵ^{Th} strains. When performing numerical analysis of laser-welded lap joints, it is imperative to define the mechanical interaction between the materials in the assembly. As a result of these mechanical interactions, the contact area between the two elements undergoes deformation. It is important to emphasize that the boundary

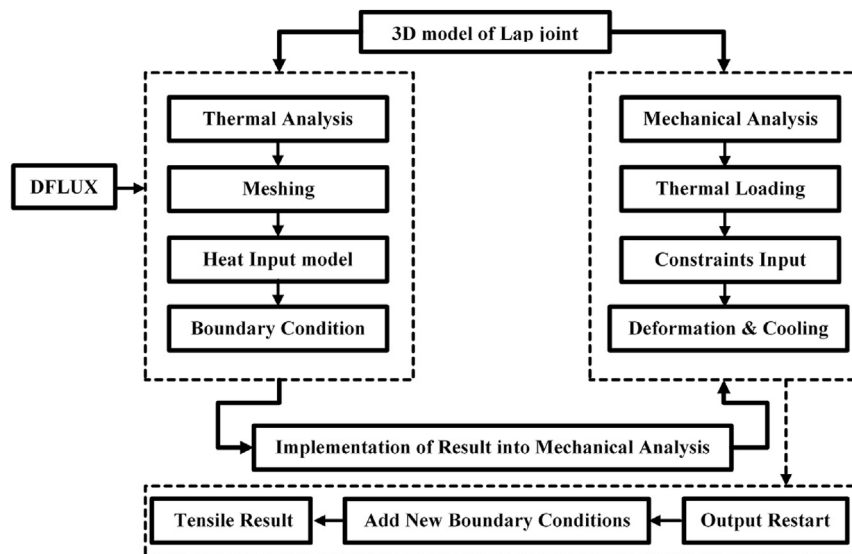
conditions applied in the mechanical analysis are prescribed to restrict rigid body movement and align with the real-world conditions observed in experiments.

3.2. Heat source modeling

The main challenge in analyzing the thermal aspects of laser welding lies in modeling the heat source. The plasma generated during the welding process heats the surface of the base metal. During penetration welding, the material absorbs the energy from the laser beam, leading to the formation of a keyhole within the base metal. In this study, a volumetric heat source with a Gaussian distribution is employed. It moves along the Z-axis and is programmed using Fortran code within the user subroutine function of the Abaqus software [19]. Thus, the controlling of the heat source can be expressed by Eq. (9) as

$$q(x, z, t) = \frac{2\eta P}{\pi r_0^2 S} e^{1-(r/r_0)^2} \left(1 - \frac{Y}{S}\right) \quad (9)$$

where q is the absorbed laser beam power, η is the assumed efficiency of laser power, r_0 is a beam radius, $r = \sqrt{x^2 + z^2}$ is the

**Fig. 4 – Diagram of FE analysis procedure.**

actual beam radius, S is penetration depth, and Y is actual penetration.

3.2.1. Discrete finite element model

The plates that are being welded together have an overlapping length of 20 mm, as illustrated in Fig. 6. For the numerical analysis model, a hexahedral element mesh shape is employed for both the thermal and mechanical analyses, with a mesh size of $25\ \mu\text{m}$ in areas where the results are of interest. This mesh size ensures optimal result quality, as it is typically smaller than the diameter of the laser beam and guarantees accurate simulation. Moreover, coarser meshes are utilized in areas that are not of particular interest, while transitional meshes are employed to connect the coarse and fine mesh regions seamlessly. The total number of finite elements used in the numerical model was 324,588, with the thermal and mechanical analyses employing C3D8T finite elements [19].

The boundary conditions applied in this analysis assume $U_x = U_y = U_z = 0$ and $U_{R_x} = U_{R_y} = U_{R_z} = 0$ to restrict linear and rotational motion concerning the x-axis, y-axis, and z-axis, as depicted in Fig. 5. The simulation process was conducted on a workstation equipped with an Intel(R) Xeon(R) CPU E5-2630 v4 @ 2.20 GHz and 192 GB RAM. The computation time for each iteration of the laser welding parameters in the thermomechanical and tensile shear force simulations was approximately 216 h.

4. Results of melting pool's shape and residual stress

4.1. Weld joint characteristics and Ni foil effect

The objective of this study was to investigate the effect of an interlayer between Ni foil and SS304 and Al6061 joints using laser penetration welding. A series of experiments were conducted to evaluate the penetration depth under different laser parameters, including power and speed, for joints with and

without Ni foil. The results, as shown in Fig. 6, revealed a strong linear relationship between the depth of penetration and the linear energy for both SS304-on-Al6061 joints with Ni foil thicknesses of 0.1 mm and 0.2 mm, as well as for joints without an interlayer. The presence of Ni foil was found to reduce heat loss from the molten stainless steel, resulting in a larger molten pool, as depicted in Figs. 6 and 7. Consequently, the penetration depth was greater in joints with Ni foil compared to those without, even when the same linear energy was applied. This suggests that the joint contact area with Ni foil is larger, leading to higher tensile shear strength compared to joints without Ni foil.

Fig. 7 illustrates the geometric features of joint cross-sections produced using laser parameters of $P = 2100\ \text{W}$ and $V = 1.5\ \text{m/min}$, where W represents the interface width, H denotes the depth of penetration in the diffusion zone, and FZ refers to the fusion line. Optical microscopy (OM) analysis in Fig. 7(a) revealed significant interfacial reactions between SS304 and Al6061, with $W1 = 583.22\ \mu\text{m}$ and $H1 = 237.55\ \mu\text{m}$ observed in the absence of an interlayer. Cross-sectional images of joints with 0.1 mm and 0.2 mm Ni foil thicknesses are presented in Fig. 7(b) and (c), respectively. It was observed that the penetration depth increased from $H2 = 257.55\ \mu\text{m}$ to $H3 = 338.95\ \mu\text{m}$, along with a corresponding increase in the interface width from $W2 = 535.89\ \mu\text{m}$ to $W3 = 775.66\ \mu\text{m}$. These findings regarding penetration depth and interface width align with the results reported by Chen et al. [3], who investigated the effect of a nickel foil interlayer on laser penetration welding of SS201 to Al5052. Their study demonstrated that the inclusion of a nickel interlayer promoted diffusion between the stainless steel and aluminum alloy, resulting in the maximum expansion of the molten aluminum pool compared to joints without a Ni foil interlayer.

4.2. Validation of the simulation model

The accuracy of the simulation model was verified by comparing it with the experimental results for the depth of

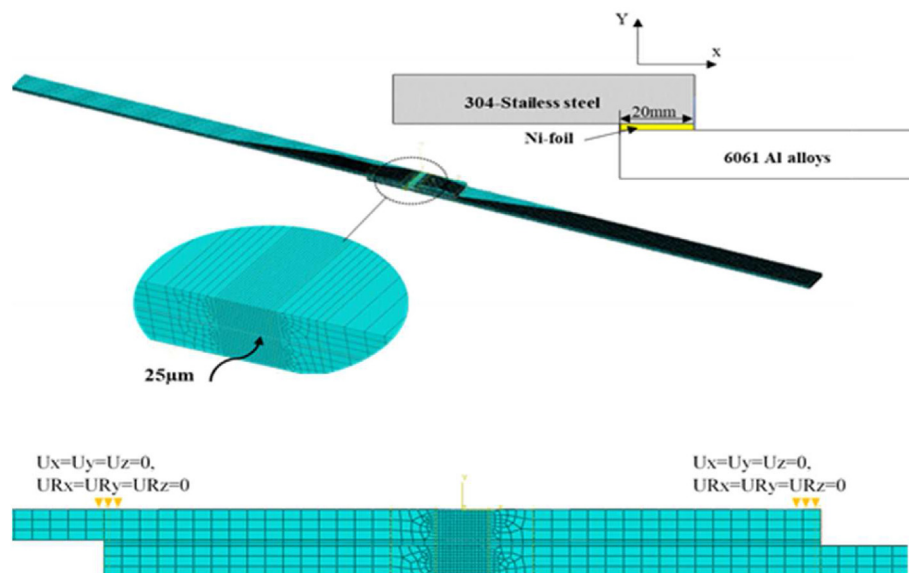


Fig. 5 – Boundary conditions and the finite element mesh.

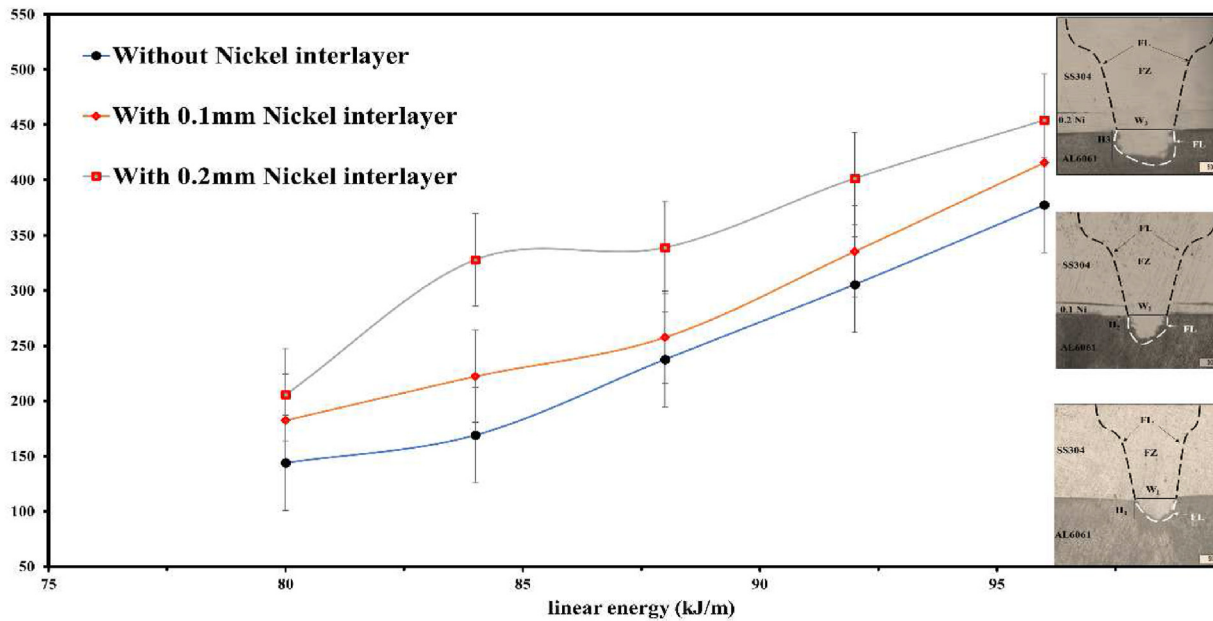


Fig. 6 – Penetration vs. linear energy for SS304-on-Al6061 with 0.1 mm and 0.2 mm Ni foil and without interlayer.

penetration and interface width of the molten pool using laser parameters of $P = 2100$ W and $V = 1.5$ m/min, which were employed in this study. The results showed that the welding process was stable when the laser beam was positioned at $Z = 5$ mm. The corresponding cross-section was selected to analyze the temperature distribution around the fusion zone, as illustrated in Fig. 8. As the thermal source moved, the temperature distribution of the components changed rapidly in both time and space. Oliveira et al. [23] emphasized the distinct characteristics of fusion welding that can be incorporated into additive manufacturing, where rapid thermal cycles are associated with small localized heat sources. Moreover, Lee et al. [24] utilized color representation in simulated images to predict the extent of the fusion zone, diffusion, and heat-affected zone in welding.

The simulation results are presented in Fig. 8, which illustrates the temperature distribution and melt pool profile of the welded joints. The joint is divided into two regions: the fusion zone, where the metals are melted and blended, and the diffusion zone, where the Ni layer is melted and diffused into the adjacent metals. Incorporating a Ni foil in the joint has an impact on the temperature during welding. The joint without Ni foil reaches a peak temperature of 2480°C , which is higher than the peak temperatures observed in joints with 0.1 mm or 0.2 mm Ni foil. The joint with a 0.1 mm Ni foil exhibits a peak temperature of 2386°C , while the joint with a 0.2 mm Ni foil exhibits a peak temperature of 2248°C . These findings suggest that utilizing a thin Ni foil can reduce the peak temperature of the joint during welding, a crucial factor in industrial applications where precise temperature control is essential.

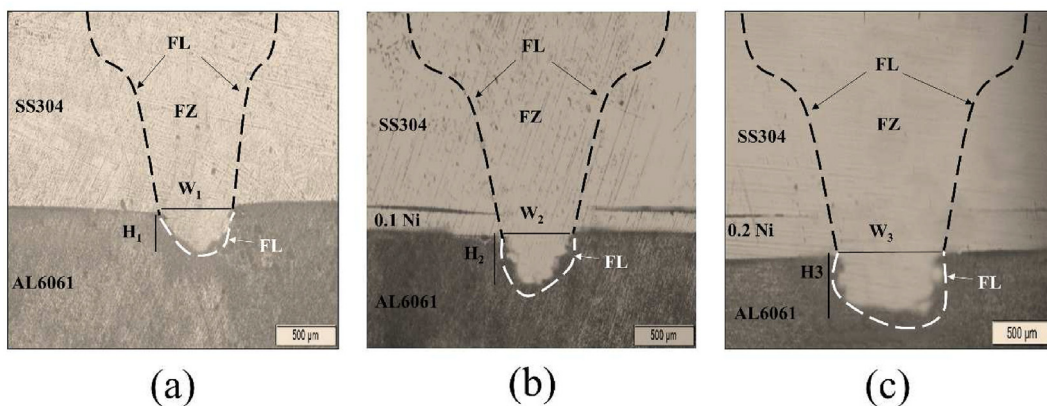


Fig. 7 – Penetration and interface at $P = 2100$ w and $V = 1.5$ m/min (a) OM image of melt pool cross-section for joint without Ni foil, (b) OM image of melt pool cross-section for a joint with 0.1 mm Ni foil, and (c) OM image of melt pool cross-section for a joint with 0.2 mm Ni foil.

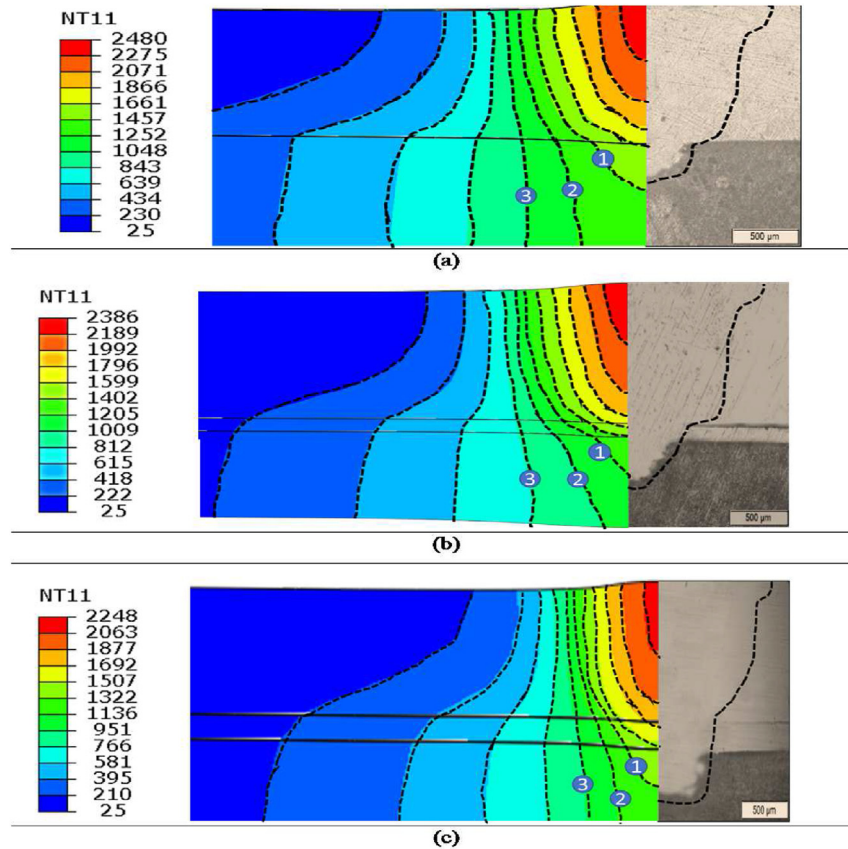


Fig. 8 – Temperature distribution and melt pool profile of simulation and experimental results for (a) cross-section for a joint without Ni foil, (b) cross-section for a joint with 0.1 mm Ni foil, and (c) cross-section for a joint with 0.2 mm Ni foil.

In Fig. 8(a), it can be observed that at Contours 2 to 3, the joint without Ni foil displays a high-temperature gradient ranging from 1252 to 1048 °C. This gradient is attributed to the high thermal conductivity of stainless steel, which acts as a heat sink and absorbs a significant amount of energy from the welding pool. Consequently, brittle Fe–Al intermetallic compounds form in the joint. However, Fig. 8(b) and (c) at Contours 2 to 3 demonstrate that the temperature gradient decreases to 1205 to 1009 °C and 951 to 766 °C for joints with 0.1 mm and 0.2 mm Ni foil, respectively. The introduction of Ni foil acts as

a thermal barrier, reducing the energy transferred to the aluminum alloy side. This reduction in temperature gradient enables the formation of a diffusion zone between the stainless steel, nickel, and aluminum. As a result, the joint becomes free from the brittle Fe–Al intermetallic compounds [25,26]. Therefore, the utilization of Ni foil significantly enhances the quality of the joint.

When a laser beam is directed at the side of an SS304 plate, it generates a molten pool within the plate, as depicted in Fig. 9(a) and (b). This process involves the absorption of a

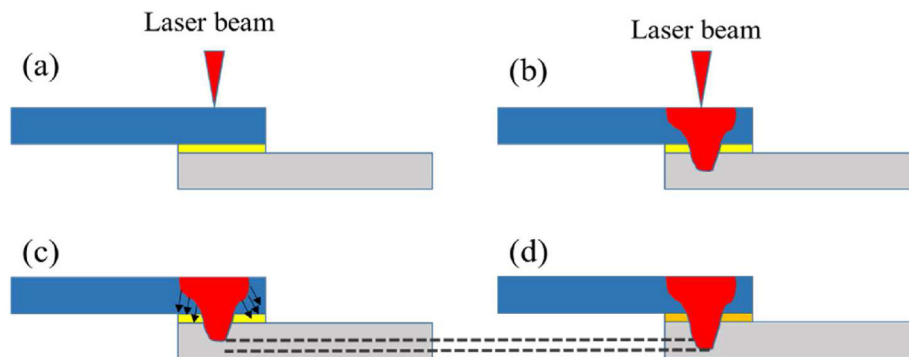


Fig. 9 – Demonstration of how the joint's physical model was created: (a) laser beam was focused on the SS plate; (b) formation of welding pool; (c) the heat was transferred from the welding pool to the SS-Ni alloy interface; (d) atomic Ni inter-diffusion at SS-Al alloy interface led to the formation of fusion and diffusion to create an intermetallic layer.

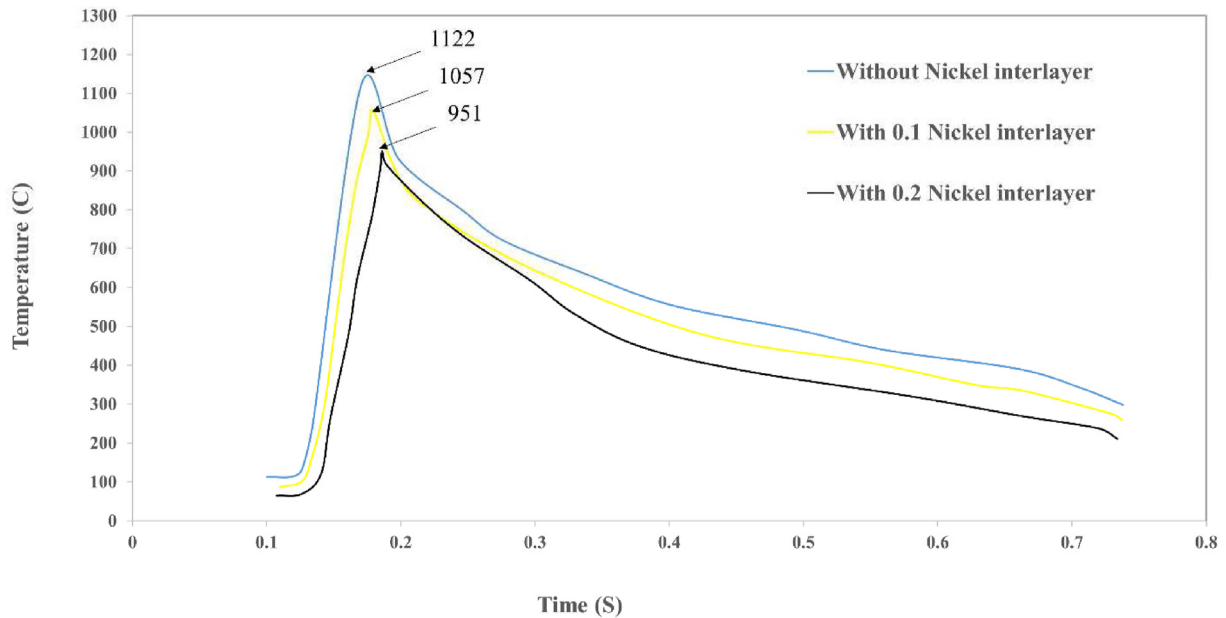


Fig. 10 – Temperature distribution on the surfaces of joints without/with Ni foil for lap welding joint.

significant amount of heat upon achieving laser penetration. The unmelted section of the SS304 plate, with its high thermal conductivity, is capable of absorbing much of this heat, allowing it to transfer heat to the Ni alloy side, as shown in Fig. 9(c). The remaining heated SS304 reaches a temperature sufficient to facilitate atomically-driven Ni inter-diffusion at the SS304–Ni alloy interface, resulting in the formation of a fusion and diffusion intermetallic layer, as observed in Fig. 9(d) (see Fig. 10).

Fig. 11(a) and (b) present a comparison between the experimental and simulated results of the weld pool's penetration depth and interface width, respectively, for laser processing parameters of 2100 W and 1.5 m/min. The simulated width and depth of the melt pool exhibit a difference of only 3.92% and 19.04%, respectively, compared to the experimental results. In other words, the computational model offers a reliable approach for predicting the shape of the weld pool in these specific weld lap joints.

4.3. Residual stress analysis of weld joints

Another primary objective of thermomechanical simulation is to capture the deformations and residual stresses of weld joints. Residual stresses in a thermomechanical simulation arise due to non-uniform plastic expansion and compression caused by heat. They can be determined once the weld joint has cooled to the same temperature as its surroundings. These stresses can be either compressive or tensile and will impact the properties of the weld differently. Most residual stresses at the surface are tensile, which can cause brittle fractures. Conversely, compressive stress at the surface will increase fatigue life [27]. Since fluid flow from the process is neglected, the accuracy of residual stress at the melting zone will be the lowest [28]. Therefore, fluid flow affects heat and mass transfer, as well as the transformation of defects in the welding pool, and the fluid flux region occurs only in a region of high temperatures with the application of laser welding.

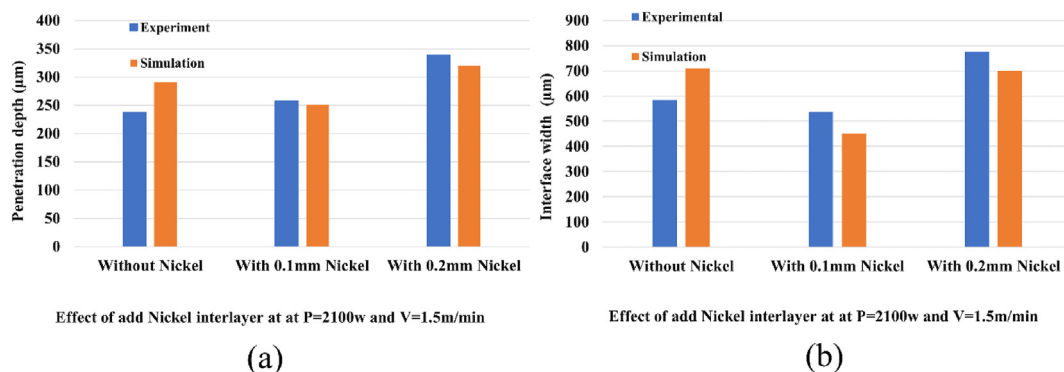


Fig. 11 – Comparison of experimental and simulation results for: (a) Penetration depth, and (b) Interface width at $P = 2100$ w and $V = 1.5$ m/min.

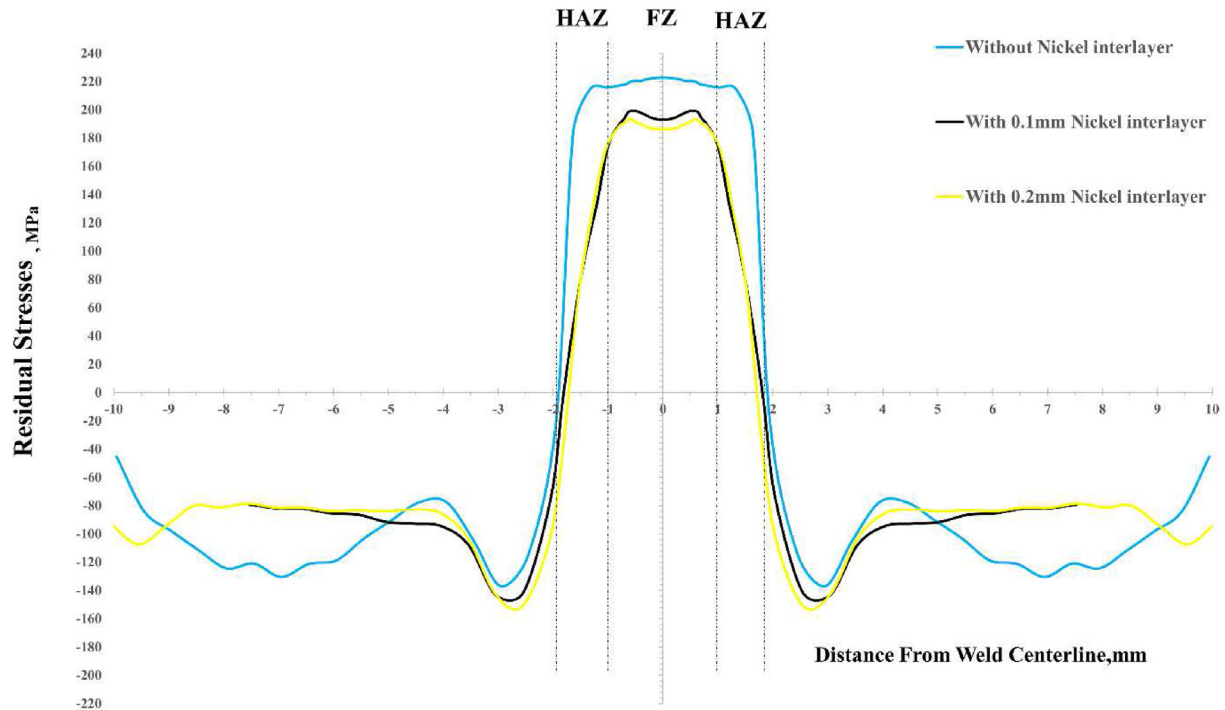


Fig. 12 – Residual stress of lap welding joints.

Fig. 12 illustrates a comparison of the impact of heat input on the heat-affected zone (HAZ) on the residual stress distribution at a distance from the weld centerline. This was achieved through finite element (FE) simulation for both a weld joint without and with a Ni foil. The results demonstrate that, for all three welding cases, the residual tensile stress is greater in the region adjacent to the weld fusion zone (FZ). It then decreases as the distance from the weld line increases, eventually approaching zero. As the heat input increases, the

residual stress in the weld gradually reduces. This is because the peak residual stresses are determined by the yield strength rather than the heat input. Consequently, the heat input does not have an impact on the residual stresses once they have been reduced to levels below the yield strength. Furthermore, the joint with the Ni foil exhibits a lower residual stress value compared to the joint without the Ni foil. This suggests that the mechanical properties of the joint are improved when the Ni foil is added (see Fig. 13).

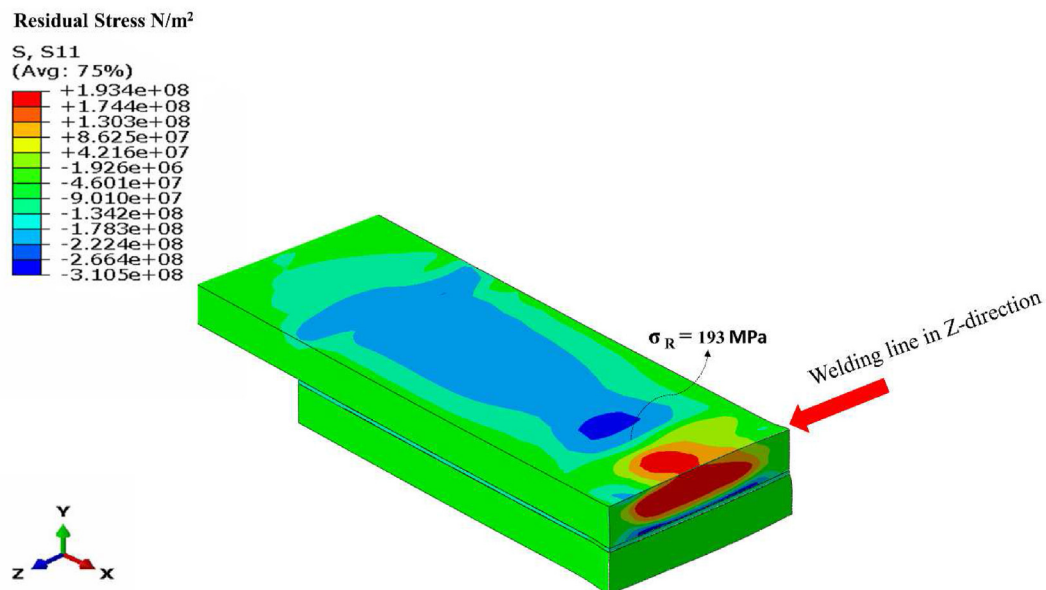


Fig. 13 – Numerically-predicted 3D residual-stress distributions for a laser power of 2100 W and a welding speed of 1.5 m/min to the case with Ni foil.

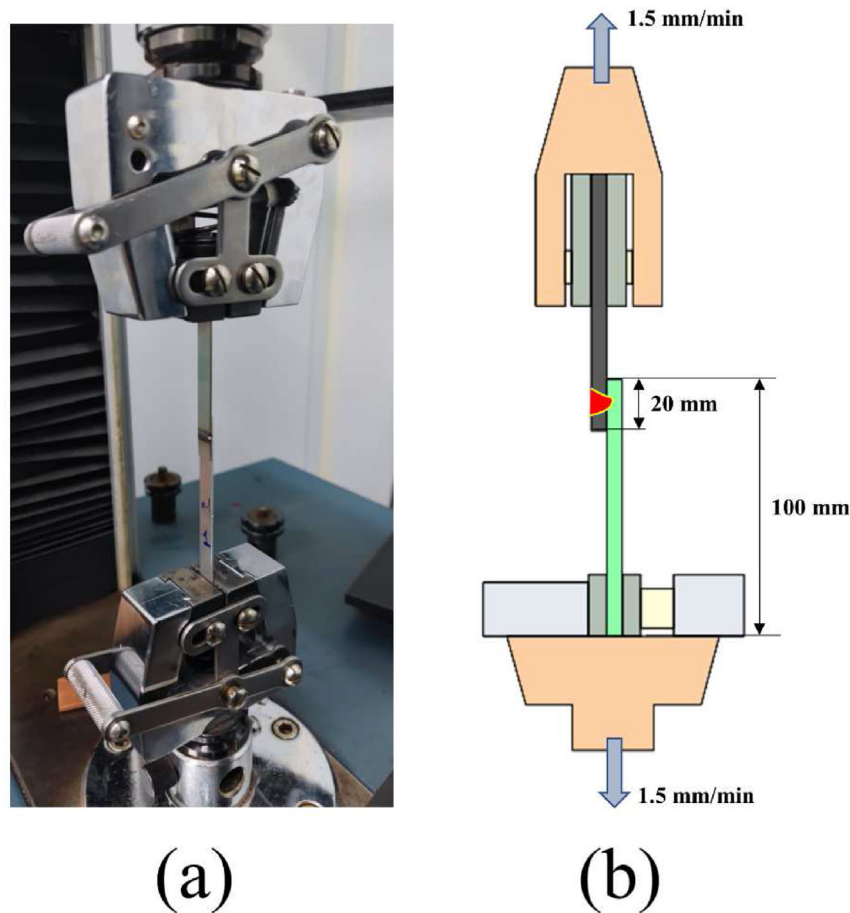


Fig. 14 – Lap-shear mechanical tests: (a) experimental setup, and (b) specimen dimensions.

5. Results of mechanical properties of lap joint for tensile shear tests

5.1. Mechanical characteristics of lap joint

Fig. 14 presents the results of a tensile shear test conducted to measure the strength of welded samples using a Shimadzu AG-I testing machine at a crosshead speed of 1.5 mm/min and a room temperature of 23 °C. To ensure repeatability and obtain the associated standard deviation, five samples were tested for each set of welding parameters, including those without Ni foil and those with 0.1 and 0.2 mm Ni foil, respectively. The gauge length used for testing was 100 mm.

Fig. 15 illustrates the mechanical properties obtained after performing tensile shear tests at a constant welding speed of 1.5 m/min and various laser powers. The results demonstrate that the highest tensile strength of the lap joint, in terms of samples without and with 0.1 mm and 0.2 mm Ni foil, respectively, was achieved when the laser power was 2100 W, as shown in Fig. 15(b), (d), and (f), with values of 813 N, 853 N, and 1026 N. The tensile shear force of the lap joints initially increased and then decreased with an increase in laser power. Fig. 15(a), (c), and (e) indicate that the corresponding maximum displacements were 1.16 mm,

1.12 mm, and 1.36 mm, respectively, based on the force-displacement diagram. It can be concluded that the tensile properties of the joint with the Ni foil interlayer are higher than those without the Ni foil interlayer [3]. The inclusion of a Ni foil interlayer can enhance the strength of the joint and limit the formation of intermetallic compounds (IMCs) at the interface of dissimilar materials (SS304 and Al6061) without an interlayer material. Furthermore, the lap joint sample with a Ni foil thickness of 0.2 mm exhibits superior strength compared to the sample with a Ni foil thickness of 0.1 mm, attributed to its greater depth of penetration and wider interface width.

5.2. Fracture shape of weld lap joint with Ni-foil

In this study, two types of fractures were observed as a result of changes in laser power and welding speed. These fractures, known as Modes I and II, occurred at the interface welding point between the nickel interlayer, Al6061, and SS304. To analyze both fracture modes, tensile shear fracture samples were used, consisting of 0.2 mm nickel foil at a laser power of 2100 W and 0.1 mm nickel foil at a laser power of 2400 W. Fig. 16(a) illustrates fracture mode I of the lap joint samples with 0.2 mm Ni foil at a laser power of 2100 W. It was observed that the fracture occurred at the bottom of the joint on the

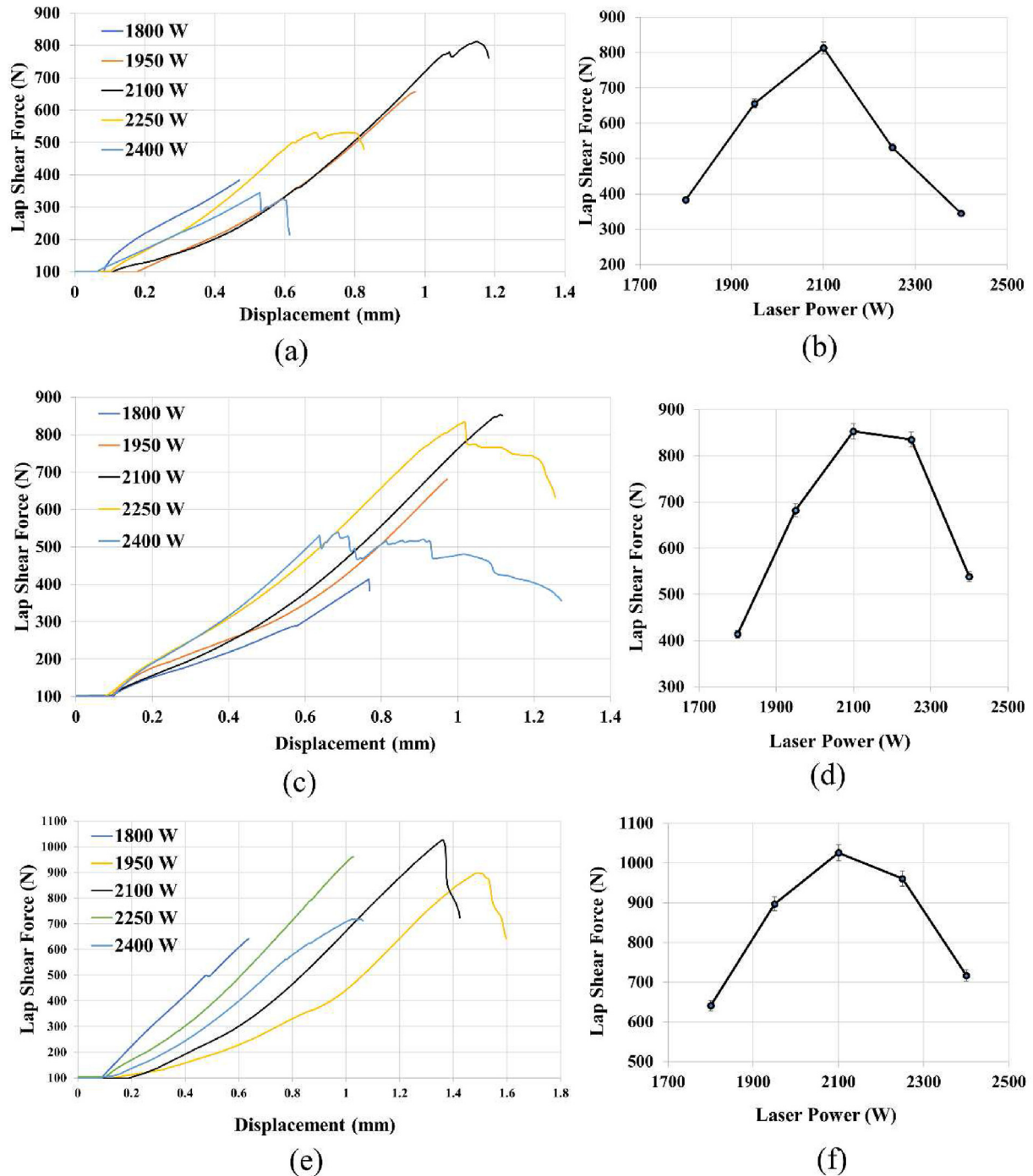


Fig. 15 – Effect of various laser powers on lap tensile shear force for (a) Samples without Ni-foil, (b) Samples without Ni-foil for average value, (c) Samples with 0.1 mm Ni-foil, (d) Samples with 0.1 mm Ni-foil for average value, (e) Samples with 0.2 mm Ni-foil, and (f) Samples with 0.2 mm Ni-foil for average value.

Al6061 side, while the SS304 and Ni foil were firmly bonded together. The strength of the fusion zone mainly relied on the mixing of the Ni foil interlayer and SS304 sheet, which had higher strength compared to the fusion zone primarily composed of the Ni foil interlayer and Al6061 sheet. This was due to the heat absorbed from the molten pool being transferred to the Ni alloy side. Fig. 16(b) shows fracture mode II of the lap joint samples with 0.1 mm Ni foil at a laser power of

2400 W. It was observed that the fracture occurred at the center of the joint on the Al6061 alloy side. Although the penetration depth of the weld was effectively increased, excessive heat input resulted in a burnt weld and a subsequent decrease in the joint's strength [2]. It is suggested that eliminating a burnt weld and its impact on joint performance can be achieved by increasing the scanning speed or focal length of laser welding.

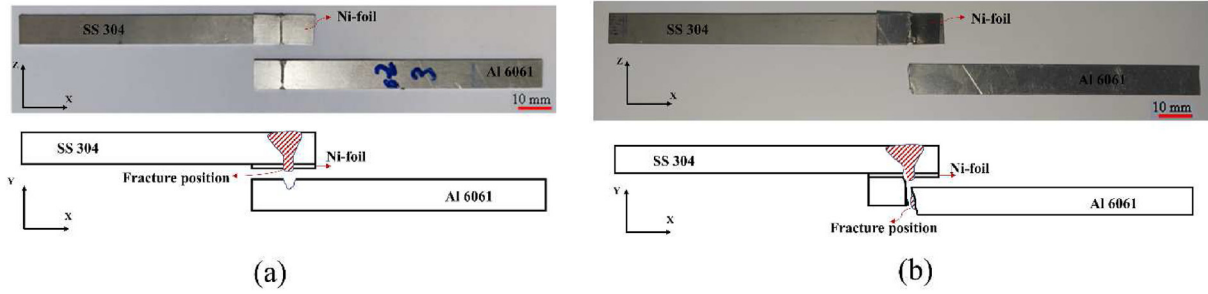


Fig. 16 – The fracture mode of samples (a) shows fracture mode I (b) shows fracture mode II.

5.3. Simulation in the tensile shear test of weld lap joint with Ni-foil

One of the goals of this study was to calculate the weld deformation induced by shear loading and to compare the experimental results with the simulation results. As illustrated in Fig. 17, the shear tensile load with a speed of 1.5 mm/min in the simulation was applied in the left and right directions of the X-axis, while the Y and Z directions were fixed. Based on the Johnson-Cook method [21], a general material model approach was chosen for the joint with 0.2 mm Nickel foil as

$$\sigma_f = \left[A + B(\epsilon^{pl})^n \right] \left[1 + C \ln \left(\frac{\dot{\epsilon}^{pl}}{\dot{\epsilon}_0^{pl}} \right) \right] \left[1 - \left(\frac{T - T_t}{T_m - T_t} \right)^m \right] \quad (10)$$

where A, B, C, n, and m are phenomenological material parameters for SS304, Ni, and Al6061, respectively, and T_t , T_m are the melting and the transition temperature. The material properties of the weld seam section are described by the Johnson-Cook parameters for the cooled-down state following welding. The identification of these parameters was based on shear tests conducted on the overlap joints.

In Fig. 17, it is demonstrated that the maximum Mises stress calculated was 272 MPa. This calculation was based on the stress distributions of the upper and lower lap plates after the addition of 0.2 mm Ni foil, which resulted in the highest achievable tensile strength. Subsequently, a shear tensile load (1.5 mm/min) was applied. During the cyclic loading process, the initial residual stress tends to relax. The extent of this relaxation is determined by the plastic strain induced by the increasing applied load. When the applied tensile mean stress is sufficiently high, the effect of the residual tensile stresses typically diminishes [29,30].

The assumption of iso-strain for the rule of mixing is strongly supported by the results of finite element simulation presented in Fig. 18(a) and (b), which depict the strain distribution in the loading direction before and after the application of the load, respectively. Furthermore, Fig. 18 illustrates a 3D model used to calibrate the Johnson-Cook parameters for the rising force-displacement curves. The results of the finite element simulation show good agreement with the experimental data, with only a 6.042% deviation from the experimental results. Moreover, the response of the weld specimen in the experimental tensile test aligns well with the plastic behavior modeling, which represents the overall behavior of the combined material.

6. Results of electron dispersive spectroscopy (EDS) analysis

Initially, the EDS analysis of the joint without a Ni-foil interlayer was examined to understand the impact of the Ni-foil interlayer on the joint's microstructures (see Fig. 19). This examination was conducted at a laser power of 2100 W and a welding speed of 3 m/min, represented by Sample No.3. Fig. 20(a) presents the SEM/EDS analysis results, demonstrating the integration of melted Al6061 with SS304. In the fusion zone (FZ), no intermetallic compound (IMC) formation was observed. The interface zone between SS304 and Al6061 contained a minimal presence of diffused, light particles, as shown in Fig. 20(a). The interfacial formations mainly consisted of FeAl/FeAl₃ [7,31]. The EDS analysis was performed at the locations of Spots 1–6, and the results are shown in Fig. 20(b). The Al6061 content slightly increased due to element diffusion, as observed at Spots 1 and 6 in the fusion zone composed of SS304. Spot 2 exhibited an Al6061 content

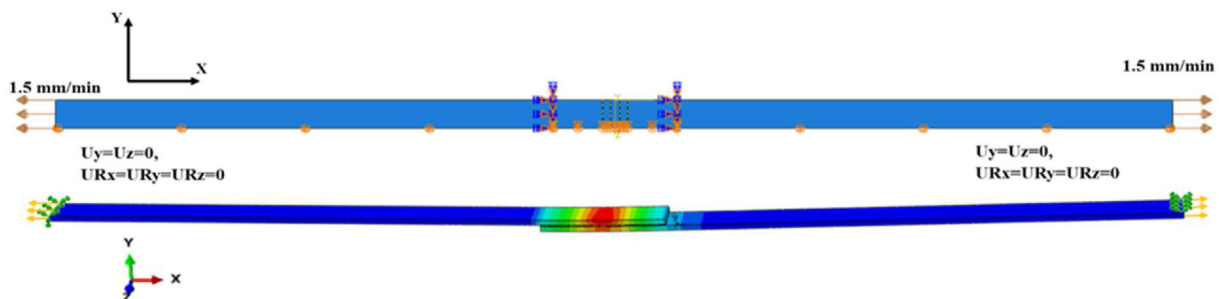


Fig. 17 – FE Model for shear test.

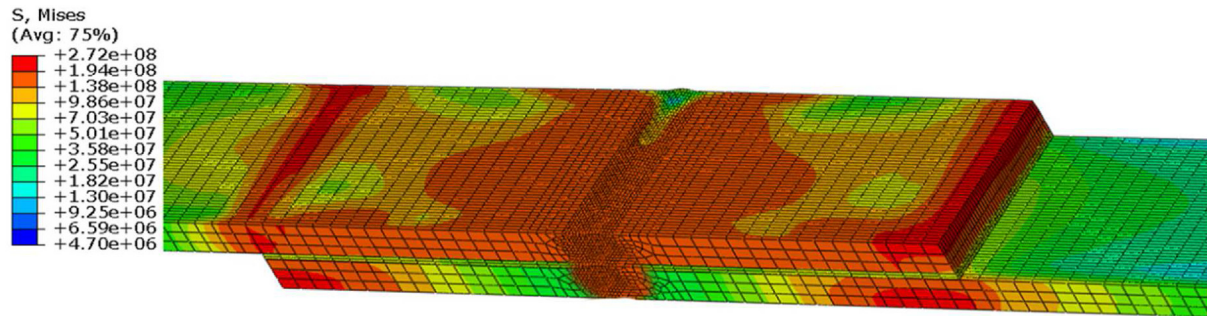


Fig. 18 – Stress distribution results after tensile load.

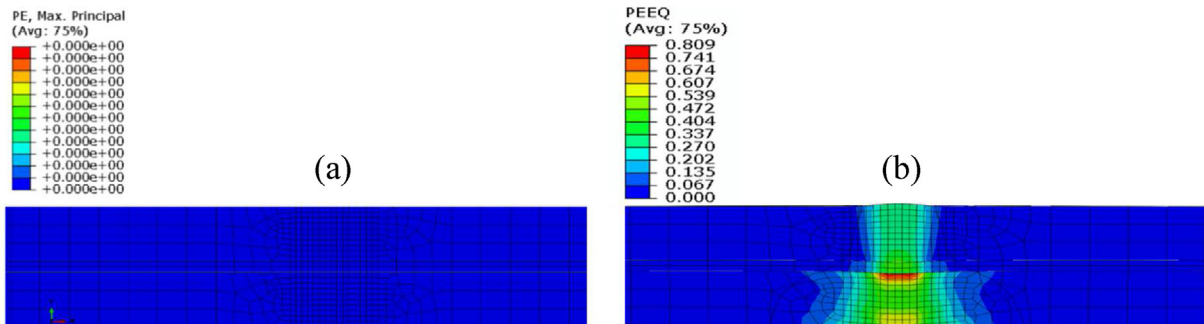


Fig. 19 – Equivalent plastic strain: (a) Before Loading; (b) After Loading.

close to 65 wt%, indicating the presence of FeAl₃. Specifically, if the Al6061 content was around 50 wt%, it was determined that the layer consisted of FeAl, as Ni atoms can substitute for Fe atoms in the lattice [32]. As can be observed at Spots 3, 4,

and 5, SS304 was still present despite a significant increase in the amount of Al due to strong diffusion. The EDS mapping results in Fig. 20(c) reveal an even distribution of Al, Fe, and Ni elements in the melt pool. Consequently, the intermetallic

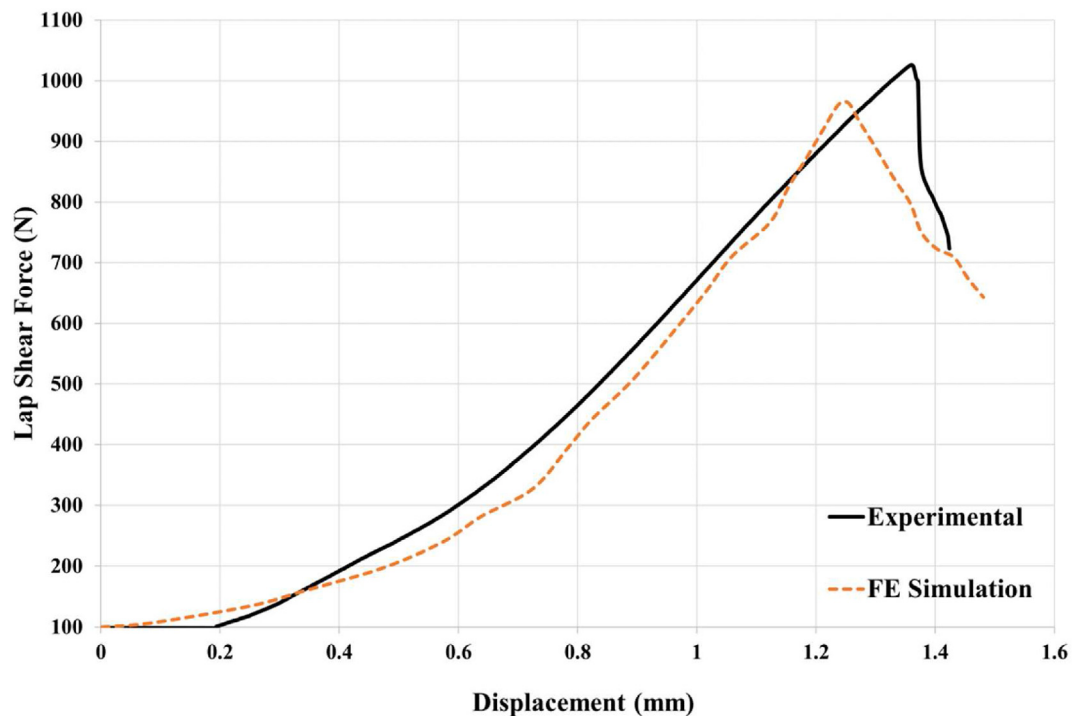


Fig. 20 – Comparison of load displacement of the experimental and numerical tensile test.

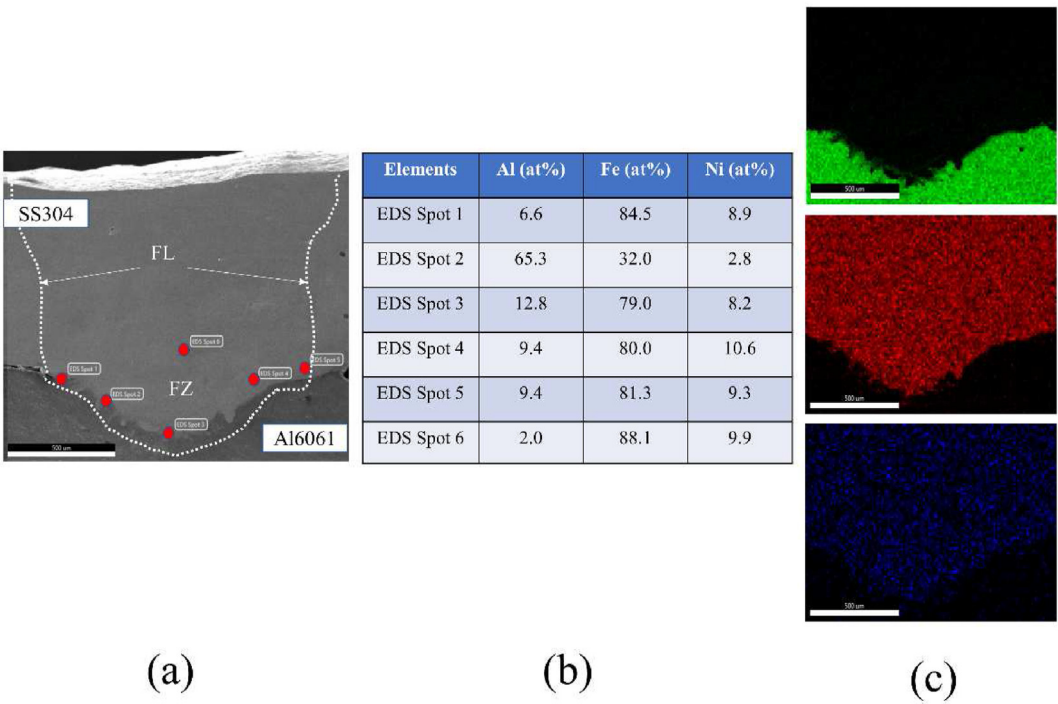


Fig. 21 – (a) EDS scanning Spots in the weld pool of the joint without a Ni interlayer, (b) EDS phase compositions at selected scanning spots, and (c) Al, Fe, and Ni elements distributions in the weld pool.

compounds are composed of two layers: one made of FeAl and the other of FeAl₃ [3,7,28].

Fig. 21(a) illustrates the results of SEM/EDS analysis conducted on the joint with a 0.2 mm Ni-foil interlayer, using a laser power of 2100 W and a welding speed of 3 m/min

(Sample No.3). The microstructure appeared to be the same as in the joint without a Ni-foil interlayer, but the molten pool of Al6061 appears to have increased, as observed in Fig. 21(a) and (b), which show the results of EDS analysis performed at Spots 1 to 8. Expansion in the molten pool and the Ni-foil is observed

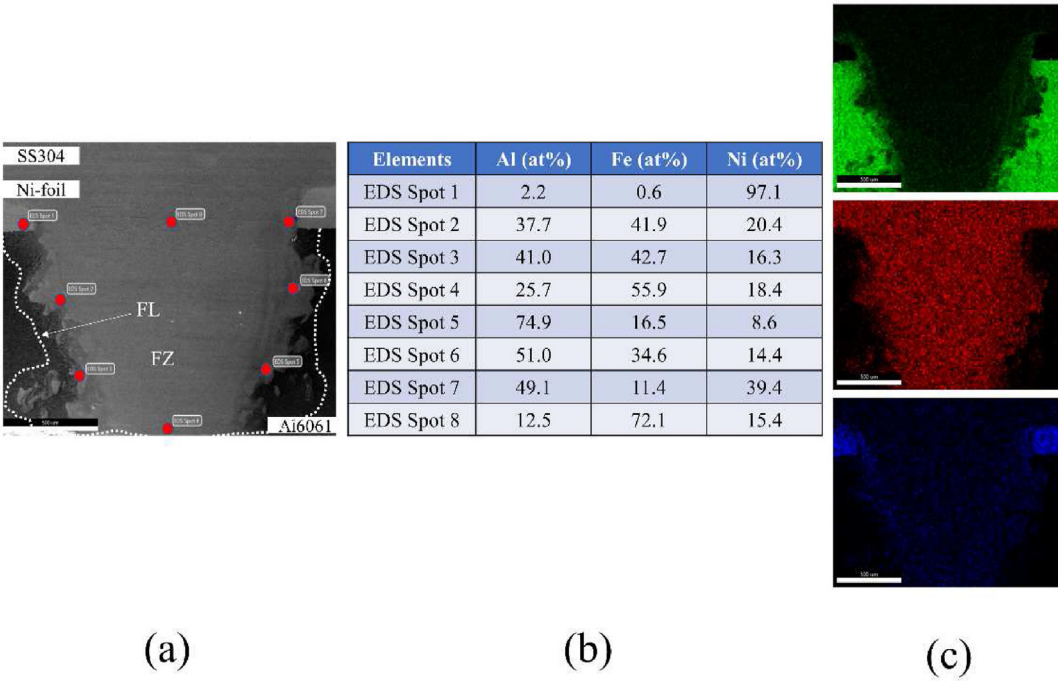


Fig. 22 – (a) EDS scanning spots in the weld pool of the joint with a Ni interlayer, (b) EDS phase compositions at selected scanning spots, and (c) Al, Fe, and Ni elements distributions in the weld pool.

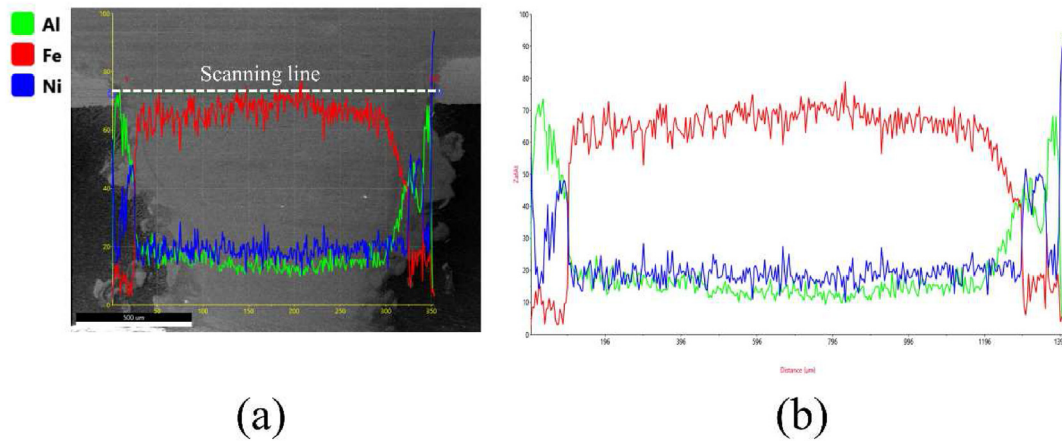


Fig. 23 – (a) EDS scanning position line, (b) Distributions of the elements as scanned with an EDS line.

at Spots 1 and 7, resulting in the formation of intermetallic compounds (IMCs) as a result of interfacial reactions. Various banded structures also appear over the reaction zone. The atomic composition at Spot 8 is close to that of SS304, indicating that the phase compositions in the fusion zone do not change significantly. In contrast, the Al/(Ni + Fe) ratio at Spots 3, 6, and 7 is quite similar to the Al6061 and Ni ratio of the IMC, which may constitute $\text{Al}_{0.9}\text{Ni}_{1.1}$ [3]. Atomic compositions at Spot 8 consist of almost 75 wt% Al6061, resulting in a layer of FeAl_3 near the Al alloy. Due to the similar concentrations of Fe and Ni, and with the Al6061 ratio at Spot 4 being close to 25 wt %, the phase compositions cannot be determined solely based on the Fe–Al or Ni–Al phase diagram. The compositions at Spot 4 are estimated to fall within the $\beta(\text{Ni}, \text{Fe})\text{Al}$ phase zone [3], as per the partial liquidus surface and vertical section of the Al–Fe–Ni phase diagram [29]. Furthermore, based on the results of the EDS analysis, it is assumed that the IMC at Spot 5 is NiAl_3 . The compositions at Spot 2 fall within the $\beta(\text{Ni}, \text{Fe})\text{Al}$ phase, based on the isothermal section of the (Fe–Al–Ni) phase diagram at 1150 °C [29]. The banded structures are formed by the $\beta(\text{Ni}, \text{Fe})\text{Al}$ phase [3].

Fig. 22(a) and (b) present the results of EDS linear scanning, which was used to examine the distribution of elements within the joint. The scanning line was utilized to investigate the interfacial reaction and the elemental distribution of the banded formations. Fig. 22(b) shows a significant increase in the Ni content in the reaction zone. This indicates the formation of a Ni-rich zone at the interfacial region during welding. The presence of the Ni-rich zone enhances the tensile performance of the joint. The EDS mapping results in Fig. 22(c) reveal a uniform distribution of the Al, Fe, and Ni elements in the melt pool. As a result, the intermetallic compounds (IMCs) with Ni foil consist of two layers: one layer composed of FeAl_3 [3,30] and the other layer composed of $\text{Al}_{0.9}\text{Ni}_{1.1}$ [3]. Fig. 23(a) illustrates the complex microstructure and numerous cracks observed in the region without Ni-foil. In contrast, Fig. 23(b) demonstrates that even in regions with complex microstructures, only a few cracks are present in joints with Ni-foil. This suggests that joints with Ni-foil have a greater ability to prevent the initiation of cracks caused by thermal expansion mismatch compared to joints without Ni-foil [3,30] (see Fig. 24).

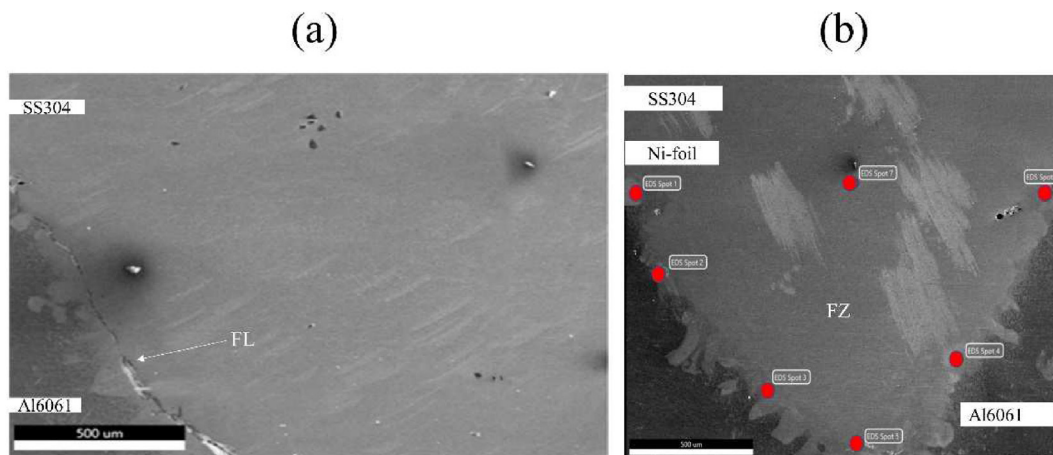


Fig. 24 – Complex microstructure cracks: (a) without Ni-foil and (b) with Ni-foil.

In summary, it can be observed that the addition of the Ni interlayer can serve as a diffusion barrier, preventing the diffusion of atoms between the two materials being bonded together. This helps to reduce the formation of brittle intermetallic compounds at the interface, which are more prone to crack formation, as depicted in Fig. 23. Furthermore, the Ni interlayer may act as a stress buffer layer, absorbing some of the stresses that would otherwise be transferred between the two materials. This helps to decrease the overall stress levels in the joint, reducing the likelihood of cracking under load. Lastly, the Ni interlayer may also have a positive effect on the microstructure of the joint. It can promote the formation of a more uniform and homogeneous microstructure, as shown in Fig. 20, thereby reducing the presence of defects and enhancing the overall mechanical properties of the joint. Overall, the addition of a Ni interlayer can have a multifaceted impact on crack inhibition by influencing diffusion, stress distribution, and microstructure.

7. Conclusions

This study has presented a numerical model for thermo-mechanical simulation of laser welding on dissimilar materials for an overlap joint with or without a nickel interlayer material. The study observed the EDS analysis and mechanical strength of welding joints, and the simulation results were validated using FE simulation, which was compared to experimental data of lap laser welded joints. The study's primary contributions and conclusions are summarised as follows.

- A thermomechanical simulation accurately predicts the width and depth of SS304–Ni–Al6061 joints, with deviations of only 3.92% and 19.04% from experimental results, respectively.
- Residual stress in the fusion weld line region is higher and decreases with distance from the weld line. The addition of a nickel foil reduces residual stress in the heat-affected zone (HAZ) and improves joint fatigue life.
- The incorporation of 0.1 and 0.2 mm Ni-foil enhances joint strength, suppresses the formation of brittle intermetallic compounds, and results in a maximum lap shear force of 1026 N.
- The simulation results for the shear load-displacement curve of the finite element (FE) model closely correspond to experimental data, with only a 6% deviation.
- Joints with Ni-foil exhibit a greater ability to prevent crack formation caused by thermal expansion mismatch compared to joints without Ni-foil.
- The Ni interlayer serves as a diffusion barrier, and stress buffer layer, and promotes a more uniform microstructure, thereby reducing the likelihood of crack formation in bonded materials.

Future studies in this field may concentrate on investigating the effects of different nickel foil interlayer thicknesses on the mechanical properties and fatigue life of the joint. Furthermore, exploring the influence of various welding parameters, including welding speed and power, on the

formation of brittle intermetallic compounds and the resulting joint strength, could provide valuable insights. The utilization of advanced characterization techniques such as electron microscopy and X-ray diffraction could also enhance the understanding of the microstructural alterations and phase transformations taking place in the joint during welding.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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REFERENCES

- [1] Shaikh H, Anita T, Dayal RK, Khatak HS. Effect of metallurgical variables on the stress corrosion crack growth behaviour of AISI type 316LN stainless steel. *Corrosion Sci* 2010;52(4):1146–54.
- [2] Ge F, Peng B, Oliveira JP, Ke W, Teshome FB, Li Y, et al. Dissimilar laser welding of a NiTi shape memory alloy to Ti2AlNb. *Metals* 2021;11(10):1578.
- [3] Chen S, Huang J, Ma K, Zhang H, Zhao X. Influence of a Ni-foil interlayer on Fe/Al dissimilar joint by laser penetration welding. *Mater Lett* 2012;79:296–9.
- [4] Chen H-C, Pinkerton AJ, Li L, Liu Z, Mistry AT. Gap-free fibre laser welding of Zn-coated steel on Al alloy for light-weight automotive applications. *Mater Des* 2011;32(2):495–504.
- [5] Pan L, He L, Niu Z, Xiao P, Zhou W, Li Y. Corrosion behavior of ytterbium hafnate exposed to water-vapor with Al (OH) 3 impurities. *J Eur Ceram Soc* 2023;43(2):612–20.
- [6] He L, Pan L, Zhou W, Niu Z, Chen X, Chen M, et al. Thermal corrosion behavior of Yb4Hf3O12 ceramics exposed to calcium-ferrum-alumina-silicate (CFAS) at 1400° C. *J Eur Ceram Soc* 2023;43(9):4114–23.
- [7] Xu T, Zhou S, Wu H, Hong C, Ma X. Effect of nickel interlayer thickness on lap joint laser welding for aluminium-steel dissimilar materials. *Sci Technol Weld Join* 2022;27(3):166–75.
- [8] Zhou D, Liu J, Lu Y, Xu S. Effect of adding powder on joint properties of laser penetration welding for dual phase steel and aluminum alloy. *Opt Laser Technol* 2017;94:171–9.
- [9] Yang J, Chen J, Zhao W, Zhang P, Yu Z, Li Y, et al. Diode laser welding/brazing of aluminum alloy to steel using a nickel coating. *Appl Sci* 2018;8(6):922.
- [10] Dilthey U. Prospects by combining and coupling laser beam and arc welding processes. *IIW Doc.*; 1999. XII–1565.
- [11] Chen Y, Lei Z, Li L, Wu L. Experimental study on welding characteristics of CO2 laser TIG hybrid welding process. *Sci Technol Weld Join* 2006;11(4):403–11.

- [12] Adak M, Mandal N. Numerical and experimental study of mitigation of welding distortion. *Appl Math Model* 2010;34(1):146–58.
- [13] Goldak JA, Akhlaghi M. *Computational welding mechanics*. Springer Science & Business Media; 2005.
- [14] Fazel-Najafabadi M, Kashani-Bozorg S, Zarei-Hanzaki A. Joining of CP-Ti to 304 stainless steel using friction stir welding technique. *Mater Des* 2010;31(10):4800–7.
- [15] Sun J, Liu X, Tong Y, Deng D. A comparative study on welding temperature fields, residual stress distributions and deformations induced by laser beam welding and CO₂ gas arc welding. *Mater Des* 2014;63:519–30.
- [16] Ranatowski E. Thermal modelling of laser welding Part I: the physical basis of laser welding. *Adv Mater Sci* 2003;3(1 (3)): 34–40.
- [17] Rothman M. Consulting editor high-temperature property data ferrous alloys, Iron—Nickel 11. 53–11. 54. 1988. ASM International Metal Park.
- [18] Ho CY, Powell RW, Liley PE. Thermal conductivity of the elements. *J Phys Chem Ref Data* 1972;1(2):279–421.
- [19] Abaqus G. Abaqus 6.11, vol. 3. Providence, RI, USA: Dassault Systemes Simulia Corporation; 2011.
- [20] Piekarska W, Kubiak M, Saternus Z. Application of Abaqus to analysis of the temperature field in elements heated by moving heat sources. *Arch Foundry Eng* 2010;10(4):177–82.
- [21] Piekarska W, Kubiak M, Saternus Z. Numerical simulation of deformations in T-joint welded by the laser beam. *Arch Metall Mater* 2013;58:1391–6.
- [22] Bokota A, Domański T. Modelling and numerical analysis of hardening phenomena of tools steel elements. *Arch Metall Mater* 2009;54(3):575–87.
- [23] Oliveira J, Santos T, Miranda R. Revisiting fundamental welding concepts to improve additive manufacturing: from theory to practice. *Prog Mater Sci* 2020;107:100590.
- [24] Lee HT, Te Chen C, Wu JL. Numerical and experimental investigation into effect of temperature field on sensitization of Alloy 690 butt welds fabricated by gas tungsten arc welding and laser beam welding. *J Mater Process Technol* 2010;210(12):1636–45.
- [25] Zhang Y, Bi Y, Zhou J, Sun D, Gu X. Butt laser welding of TC4 Titanium alloy and 304 stainless steel with Ag-base filler metal based on a hybrid connection mechanism. *Opt Laser Technol* 2020;124:105957.
- [26] Zhang Y, Chen Y, Zhou J, Sun D, Li H. Experimental and numerical study on microstructure and mechanical properties for laser welding-brazing of TC4 Titanium alloy and 304 stainless steel with Cu-base filler metal. *J Mater Res Technol* 2020;9(1):465–77.
- [27] Macdonald K. *Fracture and fatigue of welded joints and structures*. Elsevier; 2011.
- [28] Zhang W, Jiang W, Zhao X, Tu S-T. Fatigue life of a dissimilar welded joint considering the weld residual stress: experimental and finite element simulation. *Int J Fatig* 2018;109:182–90.
- [29] Ngoula DT, Beier HT, Vormwald M. Fatigue crack growth in cruciform welded joints: influence of residual stresses and of the weld toe geometry. *Int J Fatig* 2017;101:253–62.
- [30] Torkamany M, Tahamtan S, Sabbaghzadeh J. Dissimilar welding of carbon steel to 5754 aluminum alloy by Nd: YAG pulsed laser. *Mater Des* 2010;31(1):458–65.
- [31] Wang X, Chen X, Weng W, Nagaumi H, Zhou J. Effect of nickel foil thickness on microstructure and microhardness of steel/aluminium alloy dissimilar laser welding joints. *Light Metals* 2019 2019:385–93. Springer.
- [32] Rosalbino F, Scavino G, Grande MA. Electrocatalytic activity of Ni–Fe–M (M= Cr, Mn, Cu) sintered electrodes for hydrogen evolution reaction in alkaline solution. *J Electroanal Chem* 2013;694:114–21.